Database Project Synopsis

Batch No: A2

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Project Title: Smart Parking Vehicle Management System

The Smart Parking Vehicle Management System is designed to address the growing problem of urban parking congestion by providing an automated, database-driven platform for managing parking slots, vehicles, and payments. The system allows drivers to check real-time availability of parking spaces, reserve slots in advance, and make digital payments seamlessly through a unified interface, similar to modern smart city infrastructure.

The platform supports multiple parking zones or lots, each having a defined number of slots categorized by vehicle type (two-wheeler, four-wheeler, heavy vehicle). Users can register their vehicles, view available slots in their preferred zone, and book a slot for a specific duration. Upon entry, the system records the vehicle entry timestamp, and upon exit, it automatically calculates the parking fee based on duration and vehicle type.

The system ensures data integrity and consistency using DBMS concepts such as transactions, foreign key constraints, and normalization. Before confirming a booking, the system verifies slot availability, validates user credentials, and confirms payment. If a transaction fails due to a slot conflict or payment issue, the booking is rolled back and the user is notified. All parking records, including entry/exit times, fee details, and payment status, are maintained for reporting and analytics purposes.

This project demonstrates the practical application of Database Management System concepts including ER modelling, relational schema design, normalization, indexing, and transaction management. The Smart Parking Vehicle Management System integrates a responsive frontend built with React and Tailwind CSS with a robust backend powered by Node.js and Express.js, connected to a MySQL database. By simulating a real-world parking management environment, the project highlights how databases play a critical role in ensuring accuracy, reliability, and efficiency in smart city solutions.

Software Requirements

Category	Details
Operating System	Windows 10+ / Linux / macOS
Programming Language	JavaScript, SQL
Runtime	Node.js Runtime Environment
Tools / Frameworks	React.js, Tailwind CSS, Node.js, Express.js, MySQL, Visual Studio Code, Git

Hardware Requirements

Category	Details

Processor	Intel Core i3 or higher
Memory	Minimum 4 GB RAM
Other Devices	Keyboard, Mouse, Internet Connection

Guide (Incharge Faculty)	
Name(s):	
Designation:	